

Program 7: Matrix Transpose

Aim

To write a C program to find the transpose of a matrix.

Theory

The **transpose** of a matrix A of order $r \times c$ is a matrix A^T of order $c \times r$, obtained by interchanging its rows and columns.

Mathematically: $T[j][i] = A[i][j]$. The first row of A becomes the first column of the transpose, the second row becomes the second column, and so on.

Properties: $(A^T)^T = A$. A square matrix for which $A = A^T$ is called **symmetric**. Time complexity: $O(r \times c)$.

Algorithm

1. Read the number of rows r and columns c , then read the matrix A.
2. For each i from 0 to $r-1$ and each j from 0 to $c-1$, set $T[j][i] = A[i][j]$.
3. Display the original matrix and the transpose (of order $c \times r$).

C Source Code

```
/*
 * Program : Matrix Transpose
 *
 * THEORY:
 * - The transpose of a matrix A (r x c) is a matrix T (c x r) obtained by
 *   interchanging rows and columns:  $T[j][i] = A[i][j]$ .
 * - The first row of A becomes the first column of T, and so on.
 * -  $(A^T)^T = A$ . A square matrix with  $A == A^T$  is called symmetric.
 * - Time complexity:  $O(r * c)$ .
 */
#include <stdio.h>
#define MAX 10

int main() {
    int A[MAX][MAX], T[MAX][MAX], r, c;

    printf("Enter number of rows and columns: ");
    scanf("%d %d", &r, &c);

    printf("Enter elements of the matrix:\n");
    for (int i = 0; i < r; i++)
        for (int j = 0; j < c; j++)
            scanf("%d", &A[i][j]);

    for (int i = 0; i < r; i++)
        for (int j = 0; j < c; j++)
            T[j][i] = A[i][j];

    printf("\nOriginal matrix (%d x %d):\n", r, c);
    for (int i = 0; i < r; i++) {
        for (int j = 0; j < c; j++) printf("%5d", A[i][j]);
        printf("\n");
    }
    printf("\nTranspose (%d x %d):\n", c, r);
    for (int i = 0; i < c; i++) {
        for (int j = 0; j < r; j++) printf("%5d", T[i][j]);
        printf("\n");
    }
    return 0;
}
```

```
}
```

Sample Input

Input: 2 x 3 matrix A = [1 2 3; 4 5 6].

Sample Output (program run with the input above)

Enter number of rows and columns: Enter elements of the matrix:

Original matrix (2 x 3):

1	2	3
4	5	6

Transpose (3 x 2):

1	4
2	5
3	6